1. Show that $1^2+2^2+3^2+\cdots+n^2 = n(n+1)(2n+1)/6$, $n \geq 1$ by Mathematical Induction

2. Use Mathematical Induction to show that $\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\cdots+\frac{1}{\sqrt{n}} > \sqrt{n}$, $n \geq 2$

3. Prove by Mathematical Induction to prove that 3^n+7^n-2 is divisible by 8, $\forall n \geq 1$

4. How many positive integers 'n' can be formed using the digits 3, 4, 4, 5, 5, 6, 7 if n has to exceed 50,00,000?

5. Find n if ${}^{n}P_{r3} : {}^{(n+1)}P_{r2} = 3/4$

6. How many bit strings of length 10 contain
   (i) Exactly 4 1's
   (ii) Atmost 4 1's
   (iii) Atleast 4 1's
   (iv) An equal no of 0's and 1's

7. Solve the recurrence relation $a_{n+1} - a_n = 3n^2 - n$; $n \geq 0$ & $a_0 = 3$

8. Solve: $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with initial conditions $a_0 = 2, a_1 = 5, a_2 = 15$.

9. Solve the recurrence relation using generating function $a_n = 4a_{n-1} - 4a_{n-2} + 4^n$; $n \geq 2$ given that $a_0 = 2, a_1 = 8$